

Samet Tenekeci

Computational Biology & Machine Learning Researcher

Izmir Institute of Technology, Turkey

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Summary

I hold a PhD in Computer Engineering and work on AI applications in computational biology. I develop machine learning and natural language processing methods for high-dimensional biomedical and multi-omics data. I have experience teaching undergraduate and graduate courses, supervising students, contributing to international grant proposals, managing research projects, and organizing scientific workshops.

Education

PhD, Computer Engineering 2019 – 2025

Izmir Institute of Technology

Thesis: [Modeling Viral Evolution with Natural Language Processing](#)

MSc, Computer Engineering 2016 – 2019

Dokuz Eylül University

Thesis: [Discovering Disease-Causing Genes by Network Analysis](#)

BSc, Computer Engineering 2009 – 2014

Izmir University

Professional Experience

Lecturer 06/2025 – Present

Izmir Institute of Technology

- Teaching undergraduate and graduate courses
- Supervising graduation projects and mentoring students
- Conducting research in Software Engineering & AI Research Group

Research Assistant 07/2018 – 06/2025

Izmir Institute of Technology

- Assisted in teaching more than 10 computer science courses
- Contributed to several projects in Data Analytics Research Group
- Led administrative tasks within the department as the RA coordinator
- Contributed to the organization of academic conferences and workshops

Software Engineer 02/2017 – 05/2018

Airties Wireless Networks

- Developed client-specific firmware features for wireless access points
- Resolved large-scale overheating issues in production

Teaching Assistant 11/2014 – 08/2016

Izmir University

- Assisted in teaching and grading core computer science courses
- Conducted laboratory sessions for undergraduate courses
- Contributed to administrative tasks within the department

Selected Publications

Full publication list: [Google Scholar](#)

1. **Tenekeci S**, Sezgin E, Tekir S. [A contrastive learning framework for efficient viral escape prediction](#). *IEEE Transactions on Computational Biology and Bioinformatics*, 2026.
2. **Tenekeci S**, Tekir S. [Identifying promoter and enhancer sequences by graph convolutional networks](#). *Computational Biology and Chemistry*, 2024.
3. **Tenekeci S**, Işık Z. [Integrative biological network analysis to identify shared genes in metabolic disorders](#). *IEEE Transactions on Computational Biology and Bioinformatics*, 2020.
4. **Tenekeci S**, Ünlü H, Gül BA, Keleş D, Küçük M, Demirörs O. [Automating software size measurement from Python code using language models](#). *Automated Software Engineering*, 2025.
5. Ünlü H, **Tenekeci S**, Kennouche DE, Demirörs O. [Automating software size measurement with language models: Insights from industrial case studies](#). *Journal of Systems and Software*, 2026.

Research Projects

MorphoMark

10/2025 – Present

- A deep learning–based tool trained on mesh and point cloud data for 3D facial landmark localization. It supports automated detection of anatomical reference points used for geometric measurements and skeletal classification in orthodontic and craniofacial analysis. I coordinate the research and development activities of PhD-, MSc-, and BSc-level students from the dentistry and computer engineering departments.

CoV-SNN

06/2023 – 06/2025

- A protein language model trained using contrastive learning on viral protein sequences, enabling computationally efficient viral escape prediction and zero-shot variant classification. I contributed to the project proposal and led the research and development activities. The project was funded by The Council of Higher Education (YÖK) and resulted in a software tool and a peer-reviewed journal article.

AI-Estimator

01/2023 – 01/2025

- A family of domain-adapted, task-specific language models for automated software analytics. The models support automated effort estimation from software requirements. I contributed to the project proposal and to the research and development efforts. The project was funded by The Scientific and Technological Research Council of Turkey (TÜBİTAK) and resulted in a software tool and multiple journal articles.

Skills

Research Areas:

Machine learning, natural language processing, multimodal data analysis, computational biology, bioinformatics, software analytics, high-performance computing (HPC)

Technical:

Python, PyTorch, TensorFlow, R, C, Git, Docker, Linux, SLURM, Hugging Face, Streamlit

Service and Community

- Organization Committee Member, IWSM Mensura Conference
- Contributor to national and international grant proposals including Horizon Europe
- Reviewer for Journal of Supercomputing and Knowledge and Information Systems
- Campus Advisor, GitHub
- Member, Data Analytics Research Group
- Member, Software Engineering & AI Research Group